

Assignment 03

Second Year B S (Honors) 2025–2026

Course Title: Math Lab II Course Code: AMTH 250

Department of Applied Mathematics, University of Dhaka

Name:

Roll No.:

Group:

Write a *FORTRAN* program to solve each of the following problems. Always use files named according to the assignment number and problem number to take input and show output. For example, for Problem Y of Assignment X, use 'in_aXqY.txt' and 'out_aXqY.txt'.

1	The gross domestic product (GDP) of a developing economy, measured in billion dollars, is recorded at five-year intervals as follows: <table><tr><th>Year</th><th>2000</th><th>2005</th><th>2010</th><th>2015</th><th>2020</th><th>2025</th></tr><tr><td>GDP (billion USD)</td><td>52.4</td><td>61.8</td><td>75.6</td><td>93.2</td><td>118.5</td><td>149.7</td></tr></table> Using Newton’s forward difference interpolation formula, estimate the GDP in the year 2007. Your FORTRAN program should construct and display the complete difference table and then calculate the required GDP.	Year	2000	2005	2010	2015	2020	2025	GDP (billion USD)	52.4	61.8	75.6	93.2	118.5	149.7
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GDP (billion USD)	52.4	61.8	75.6	93.2	118.5	149.7									
2	A commercial bank records the maturity value of a fixed deposit of Tk. 100,000 for different investment periods as follows: <table><tr><th>Investment period (years)</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr><tr><td>Maturity value (Tk.)</td><td>106500</td><td>113420</td><td>120790</td><td>128620</td><td>136950</td></tr></table> Using Newton’s backward difference interpolation formula, construct the complete difference table and display it. Then, estimate the maturity value for an investment period of 4.5 years.	Investment period (years)	1	2	3	4	5	Maturity value (Tk.)	106500	113420	120790	128620	136950		
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3	The exchange rate between the US dollar and Bangladeshi taka is observed at selected times during a trading day. <table><tr><td>Time x (hours after market opening)</td><td>0.0</td><td>1.0</td><td>2.5</td><td>4.0</td><td>6.0</td></tr><tr><td>Exchange rate f(x) (Tk./USD)</td><td>118.20</td><td>118.45</td><td>118.93</td><td>119.36</td><td>120.05</td></tr></table> Use Newton's divided difference formula to construct the interpolating polynomial of degree four. Show the complete divided difference table appropriately and hence estimate the exchange rate at x = 3.0 hours.	Time x (hours after market opening)	0.0	1.0	2.5	4.0	6.0	Exchange rate f(x) (Tk./USD)	118.20	118.45	118.93	119.36	120.05
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4	The future value index of an investment is modeled by $f(x) = 0.3x e^{0.15x^2}, \text{ where } x \text{ represents the investment period in years.}$ Using the data corresponding to $x_0 = 0.0, x_1 = 0.5, x_2 = 1.0, x_3 = 1.5, x_4 = 2.0$ apply Lagrange's interpolation formula to approximate f(1.25). Your program should: (a) Calculate the function values at the five interpolation points. (b) Determine the approximate value of f(1.25). (c) Calculate the exact value using the given function. (d) Compare the interpolated and exact values and report the absolute error.												

N.B. Students should bring the required algorithms for their convenience.